

Exchange Rates and Monetary Policy with Heterogeneous Agents: Sizing up the Real Income Channel

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- Does heterogeneity matter for transmission of international shocks?
- Study the impact of a depreciation with HANK model
 - HANK + small open economy model as in Galí and Monacelli (2005)
 - Empirically relevant MPCs
- Incomplete markets give rise to the real income channel, and heterogeneity amplifies it
- Weaker output expansion or even contractionary depreciations

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Household's problem

- The household's problem:

$$V_t(a, e) = \max_{c_F, c_H, a'} u(c_F, c_H) - v(N_t) + \beta \mathbb{E}_t [V_{t+1}(a', e')]$$

$$\text{s.t.} \quad \frac{P_{Ft}}{P_t} c_F + \frac{P_{Ht}}{P_t} c_H + a' = (1 + r_t^p) a + e \frac{W_t}{P_t} N_t$$

$$a' \geq \underline{a}$$

- CRRA utility

$$u(c_F, c_H) = \frac{c^{1-\sigma}}{1-\sigma}, \quad v(N) = \psi \frac{N^{1+\varphi}}{1+\varphi}$$

- Labor supply N_t determined by union

Consumption basket

- Two goods (home, foreign) composing the consumption basket

$$c = \left[\alpha^{1/\eta} C_F^{(\eta-1)/\eta} + (1 - \alpha)^{1/\eta} C_H^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}$$

where $1 - \alpha$ is the home bias, $\eta > 0$ is the elasticity of substitution

- The CPI is

$$P_t = [\alpha P_{Ft}^{1-\eta} + (1 - \alpha) P_{Ht}^{1-\eta}]^{1/(1-\eta)}$$

- Households have the same consumption baskets

$$c_{Ft}(a, e) = \alpha \left(\frac{P_{Ft}}{P_t} \right)^{-\eta} c_t(a, e)$$

$$c_{Ht}(a, e) = (1 - \alpha) \left(\frac{P_{Ht}}{P_t} \right)^{-\eta} c_t(a, e)$$

Exchange rate

- Normalize the price of foreign goods in foreign currency as $P_{Ft}^* = P_t^* = 1$
- The law of one price holds for imported and exported goods

$$P_{Ft} = \mathcal{E}_t$$

$$P_{Ht}^* = \frac{P_{H,t}}{\mathcal{E}_t},$$

where $\mathcal{E}_t$ is the nominal exchange rate

- The real exchange rate is

$$Q_t \equiv \frac{\mathcal{E}_t}{P_t}$$

Elasticities of imports and exports

- The aggregate import is

$$C_{Ft} = \alpha \left(\frac{P_{Ft}}{P_t} \right)^{-\eta} C_t,$$

- the elasticity wrt the relative price of imports $P_{F,t}/P_{H,t}$ is $\eta(1 - \alpha)$

- The exports are

$$C_{Ht}^* = \alpha \left(\frac{P_{Ft}}{P_t} \right)^{-\gamma} C^*$$

- γ is the elasticity of substitution across countries for foreign households
- the elasticity is γ
- Denote by χ the trade elasticity

$$\chi = \eta(1 - \alpha) + \gamma$$

Production

- Production function is

$$Y_t = ZN_t$$

- Firms are monopolistically competitive, the price is

$$P_{Ht} = \mu \frac{W_t}{Z},$$

where $\mu = \epsilon/(\epsilon - 1)$, and ϵ is the elasticity of substitution between varieties within a country

- Firms maximize firm value

$$\max \frac{P_{Ht} Y_t - W_t N_t}{P_t} + p_t$$

where the first term is dividends D_t and p_t is the end-of-period price on the shares outstanding

- Risk-neutral, unconstrained mutual fund can invest into four assets:
 - domestic nominal bonds, interest rate i
 - foreign nominal bonds, interest rate i^*
 - domestic firm shares with return $(p_{t+1} + D_{t+1})/p_t$
 - foreign firm shares
- The equality of expected returns implies the UIP condition:

$$1 + i_t = (1 + i_t^*) \frac{\mathcal{E}_{t+1}}{\mathcal{E}_t}$$

and ex-ante return on domestic stocks:

$$1 + r = \frac{p_{t+1} + D_{t+1}}{p_t}$$

- As the portfolio is indeterminate, assume invest only in domestic stocks

Unions and monetary policy

- A union employs all households for N_t hours and sets nominal wages by maximizing the welfare of the average household
 - The resulting wage Phillips curve:

$$\pi_{wt} = \kappa_w \left(\frac{v'(N_t)}{\frac{1}{\mu_w} \frac{W_t}{P_t} u'(C_t)} - 1 \right) + \beta \pi_{wt+1}$$

- Monetary rule holds the real rate constant:

$$i_t = r_{ss} + \pi_{t+1} + \epsilon_t$$

- No fiscal policy

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- The benchmark model is based on Gali and Monacelli (2005)
 - Representative agent
 - Complete markets across households and countries
 - Extended to allow for foreign discount factor shocks
- Backus-Smith condition:

$$\frac{Q_t}{\mathcal{B}_t} C_t^{-\sigma} = C_{ss}^{-\sigma}$$

where $\mathcal{B}_t = \prod_{s \geq t} \left(\frac{1+i_s^*}{1+r_{ss}} \right)$ is the exogenous preference shifter of foreign households

- Assuming $Q_\infty = 1$, from UIP can get $Q_t = \mathcal{B}_t$

Exchange rate shock in a RA model

- C_t does not respond to the exchange rate shock

$$dC_t = 0 \quad \forall t$$

- Depreciation is expansionary

$$dY_t = \frac{\alpha}{1 - \alpha} \chi dQ_t \quad \forall t$$

- Expansion is increasing in trade elasticity χ

Exchange Rate Shocks

- Home goods market clearing condition can be written as

$$Y_t = (1 - \alpha) \left(\frac{P_{Ht}}{P_t} \right)^{-\eta} C_t + \alpha \left(\frac{P_{Ht}}{\mathcal{E}_t} \right)^{-\gamma} C^*$$

- Expenditure switching effect: depreciation makes consumers substitute towards home goods
- Does domestic spending C_t change due to the income effect?

Real income channel

- Incomplete markets $\Rightarrow$ real income fluctuations matter for consumption
- Income consists of two parts:

- Real wage

$$\frac{W_t}{P_t} N_t = \frac{1}{\mu} \frac{P_{Ht}}{P_t} Y_t$$

- Asset returns r^P , pinned down by the dividend

$$D_t = \left(1 - \frac{1}{\mu}\right) \frac{P_{Ht}}{P_t} Y_t$$

- The path of consumption C_t is a function of the path of real income $\frac{P_{Ht}}{P_t} Y_t$

Real income channel

Proposition

In response to a shock to the real exchange rate dQ , the impulse response of consumption is given by

$$dC = - \underbrace{\frac{\alpha}{1-\alpha} M dQ}_{\text{Real income channel}} + \underbrace{M dY}_{\text{Multiplier}}$$

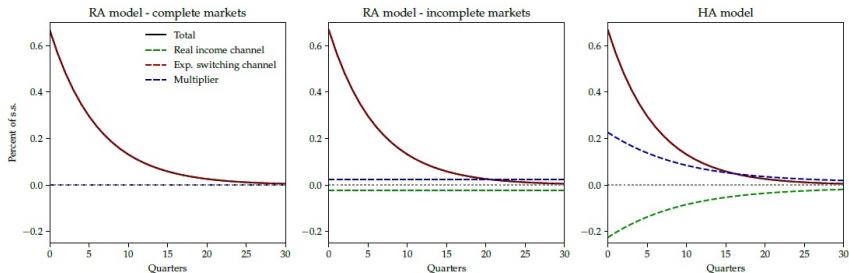
and the impulse response of output dY is determined by

$$dY = \underbrace{\frac{\alpha}{1-\alpha} \chi dQ}_{\text{Exp. switching channel}} - \underbrace{\alpha M dQ}_{\text{Real income channel}} + \underbrace{(1-\alpha) M dY}_{\text{Multiplier}}$$

Solution

where M is 'intertemporal MPC', $\mathcal{M}_{t,s} = \frac{\partial C_t}{\partial Y_s}$

FX shock when $\chi = 1$ and its transmission channels

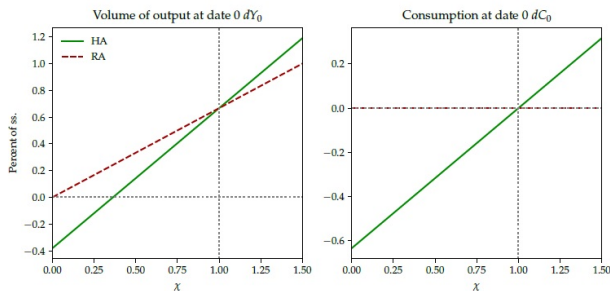


Neutrality result

Proposition

Consider any model of consumption characterized by a matrix M , and an arbitrary exchange rate shock dQ . If $\chi = 1$, all aggregate quantities and prices are the same as in the RA model, and in particular, $dY = dY^{RA}$. Moreover, provided that $M > 0$, for a depreciation shock $dQ \geq 0$, we have

$$dY \leq dY^{RA} \quad \text{and} \quad dC \leq 0 \quad \Leftrightarrow \quad \chi \leq 1$$



Contractionary depreciations

Proposition

If $\chi < 1 - \alpha$, the output response to a depreciation shock $dQ_t \geq 0$ has negative net present value, $\sum_{t=0}^{\infty} (1+r)^{-t} dY_t < 0$ in the heterogeneous-agent model. Moreover, given a depreciation shock, there is a threshold χ^* between $(1 - \alpha)M_{0,0}$ and 1 such that for any $\chi < \chi^*$, the output response is negative on impact, $dY_0 < 0$.

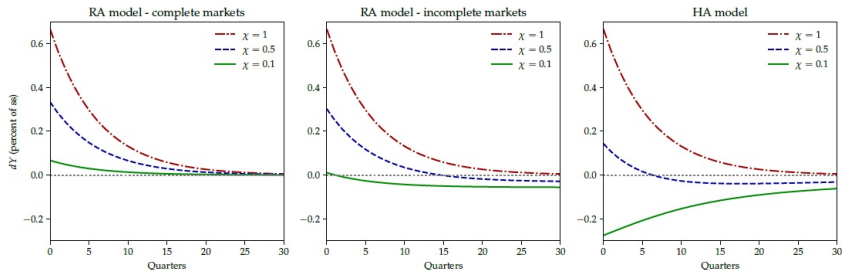


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Additional model elements

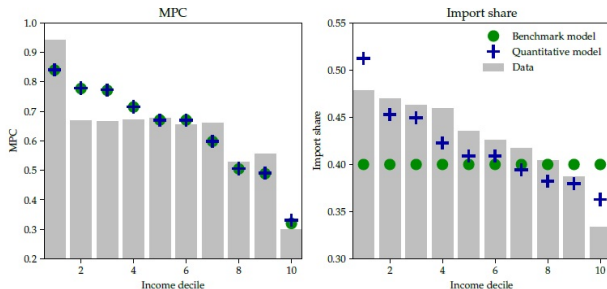
- Non-homothetic preferences

$$c = \left[\alpha^{1/\eta} (c_F - \underline{c})^{(\eta-1)/\eta} + (1 - \alpha)^{1/\eta} c_H^{(\eta-1)/\eta} \right]^{\eta/(\eta-1)}$$

- PPI inflation based Taylor rule
- Sticky prices Phillips curves
- Imperfect exchange rate pass-through
- Households only can adjust the relative consumption of different countries' goods with probability $1 - \theta$ Equations
 - Makes trade elasticity both shock- and time-dependent
 - Lower elasticity in the short run

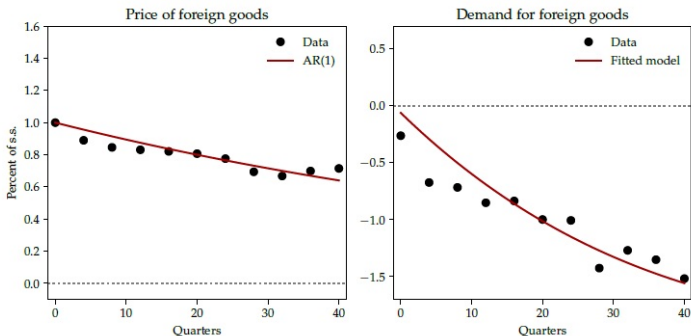
Calibration

- Calibrated for Mexico
- Targeted moments:
 - Average annual MPC, std of annual MPC (Hong 2020)
 - Average tradable share, std of tradable share (Mexican spending survey)
- Calibration outcomes:



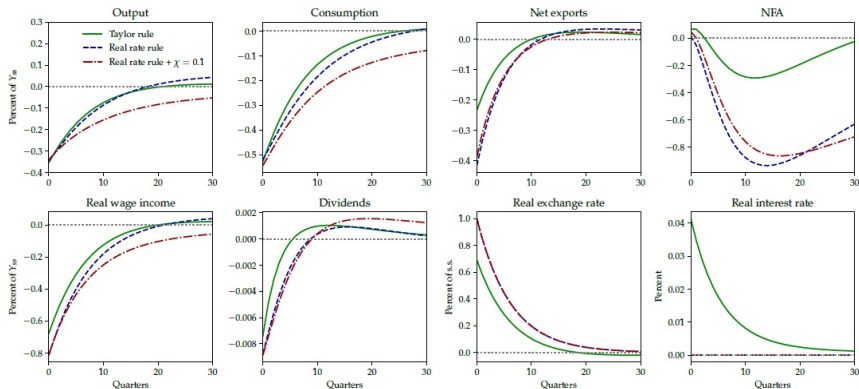
Calibration

- Delayed substitution targets the moments from Boehm, Levchenko and Pandalai-Nayar (2020)
 - Identify exogenous tariff changes and the response of trade flows
 - Used to calibrate θ
- Calibration outcomes:



All parameters

Contractory depreciations



When does the income channel matter?

	Baseline	Low α	High MPC	Full DCP	Homothetic	High ST χ	$\theta_F = 0.7$	$\theta_F = 0.9$
dY_0	- 0.35	- 0.26	- 0.39	- 0.31	- 0.32	- 0.30	- 0.09	- 0.02
$\sum_{t=0}^7 dY_t$	- 1.81	- 1.48	- 1.85	- 1.53	- 1.54	- 1.14	- 0.95	- 0.22

- High openness
- high MPC and non-homothetic preferences
- High exchange-rate pass-through into import prices

- Introduce heterogeneous households into a SOE NK model
- Real income channel can generate contractionary depreciations when it takes time to substitute away from imported goods
- Comments:
 - How well does the model fit non-targeted moments?
 - Would be interesting to show real income channel on micro-level
 - Heterogeneity in exposure to international shocks (savings, borrowing, employment sector) may matter for the size of the real income channel

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4 Appendix

Equilibrium

Given sequences of foreign interest rate shocks $\{i_t^*\}$ and monetary shocks $\{\epsilon_t\}$, an initial wealth distribution $\mathcal{D}_0(a, e)$, and an initial portfolio allocation for the mutual fund, a *competitive equilibrium* is

- a path of policies $\{c_{Ht}(a, e), c_{Ft}(a, e), c_t(a, e), a_{t+1}(a, e)\}$ for households,
- distributions $\mathcal{D}_t(a, e)$
- prices $\{\mathcal{E}_t, Q_t, P_t, P_{Ht}, P_{Ft}, W_t, p_t, i_t, r_t, r_t^p\}$
- aggregate quantities $\{C_t, C_{Ht}, C_{Ft}, Y_t, A_t, D_t, nfa_t\}$
such that all agents optimize, firms optimize, and the domestic goods market clears:

$$C_{Ht} + C_{Ht}^* = Y_t$$

Assume the long-run exchange rate returns to its steady state level,
 $Q_\infty = Q_{ss}$.

Unique output response

Proposition

Assuming $M \geq 0$, the equilibrium output response is unique and given by

$$dY = \underbrace{\left(\sum_{k \geq 0} (1 - \alpha)^k M^k \right)}_{=(I - (1 - \alpha)M)^{-1}} \left(\frac{\alpha}{1 - \alpha} \chi dQ - \alpha M dQ \right)$$

Back

Monetary policy shock

- Focus on real interest rates
- Two effects of $\{r_t\}$:
 - Given $Q_\infty = 1$ and $i^* = r_{ss}$, from UIP

$$Q_t = \prod_{s \geq t} \left(\frac{1 + r_{ss}}{1 + r_s} \right)$$

- Direct effect through asset returns as in the closed economy, the interest rate response matrix M^r , where $M_{t,s}^r = (1 + r) \frac{\partial C_t}{\partial r_s}$
- Generalized version of the international Keynesian cross:

$$dY = \underbrace{(1 - \alpha)M^r dr}_{\text{Interest rate channel}} + \underbrace{\frac{\alpha}{1 - \alpha}\chi dQ}_{\text{Exp. switching channel}} - \underbrace{\alpha M dQ}_{\text{Real income channel}} + \underbrace{(1 - \alpha)M dY}_{\text{Multiplier}}$$

$$\text{where } dQ_t = - \sum_{s \geq t} \frac{dr_s}{1 + r_{ss}}$$

Monetary policy shock

- In a RA model with log preferences, $M = 0$ and $M^r = -U$, where U is a matrix with 1 on and above diagonal, so

$$dY^{RA} = -\left((1 - \alpha) + \frac{\alpha}{1 - \alpha}\chi\right)Udr$$

- In a HA model,

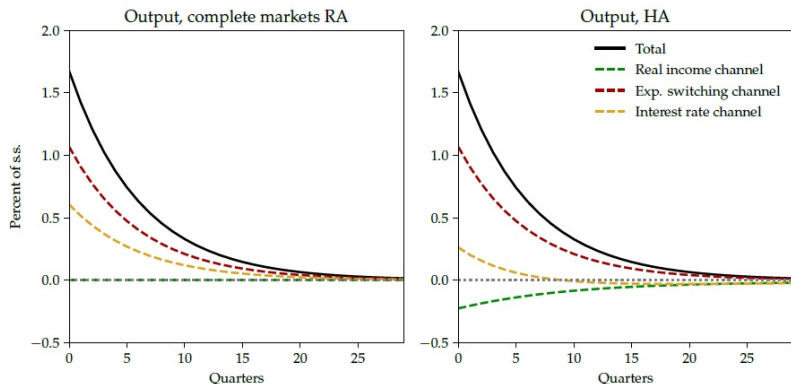
Proposition

Assume $\sigma = 1$, and consider an arbitrary first-order monetary policy shock dr . If $\chi = 2 - \alpha$, all aggregate quantities and prices are identical in HA and RA models. Moreover, provided that $M > 0$, for an accommodative shock $dr \leq 0$,

$$dY \leq dY^{RA} \quad \Leftrightarrow \quad \chi \leq 2 - \alpha$$

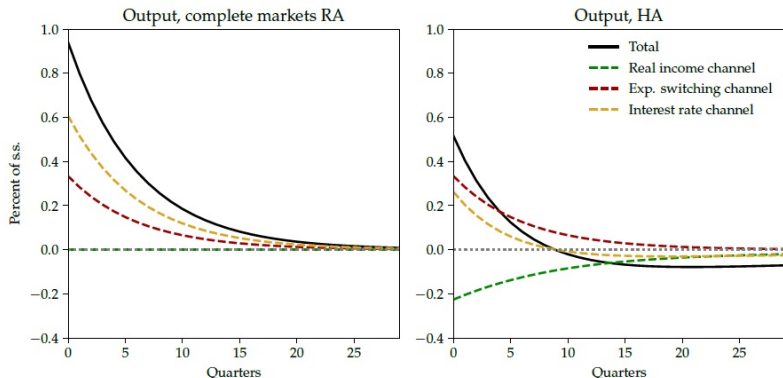
Monetary policy: neutrality result

(a) Neutral case: $\chi = 2 - \alpha$



Monetary policy: lower elasticity

(b) Lower trade elasticity: $\chi = 0.5$



Phillips curves

- Home goods:

$$\pi_{Ht} = \kappa_H \left(\mu \frac{W_t}{Z_t P_{Ht}} - 1 \right) + \frac{1}{1+r} \mathbb{E}_t [\pi_{H,t+1}]$$

- Imported goods in domestic currency:

$$\pi_{Ft} = \kappa_F \left(\mu_F \frac{\mathcal{E}_t}{P_{Ft}} - 1 \right) + \frac{1}{1+r} \mathbb{E}_t [\pi_{F,t+1}]$$

- Exported goods in domestic currency:

$$\pi_{Ht}^* = \kappa_{H^*} \left(\mu_{H^*} \frac{P_{Ht}}{\mathcal{E}_t P_{Ht}^*} - 1 \right) + \frac{1}{1+r} \mathbb{E}_t [\pi_{H,t+1}^*]$$

Delayed substitution

- Targeted ratios of domestic/foreign home good consumption to overall consumption:

$$d \log \hat{x}_{Ht} = -(1 - \beta\theta)\eta d \log \frac{P_{Ht}}{P_t} + \beta\theta d \log \hat{x}_{Ht+1}$$

$$d \log \hat{x}_{Ht}^* = -(1 - \beta\theta)\gamma d \log \frac{P_{Ht}}{\mathcal{E}_t} + \beta\theta d \log \hat{x}_{Ht+1}^*$$

- The dynamics is

$$d \log \frac{C_{Ht}}{C_t} = (1 - \theta) d \log \hat{x}_{Ht} + \theta d \log \frac{C_{Ht-1}}{C_{t-1}}$$

$$d \log C_{Ht}^* = (1 - \theta) d \log \hat{x}_{Ht}^* + \theta d \log C_{Ht-1}^*$$

Calibrated parameters

Parameter	Benchmark model	Quantitative model	Parameter	Benchmark model	Quantitative model
σ	1	1	μ	1.03	1.028
φ	2	2	s.s. nfa	0	0
η	$\frac{\{0.1, 0.5, 1, 2 - \alpha\}}{2 - \alpha}$	4	σ_e	0.6	0.6
γ	$= \eta$	$= \eta$	ρ_e	0.92	0.92
θ	n.a.	0.987	θ_w	0.95	0.95
β	0.954	0.947	θ_H	0	0.66
Δ	0.06	0.067	θ_{H^*}	0	0.66
α	0.4	0.323	θ_F	0	0
$\underline{c}$	0	0.114	ϕ	n.a.	1.5

	Moment	Data	Benchmark model	Quantitative Model
	Average annual MPC	0.632	0.636	0.637
	Std of annual MPC	0.152	0.151	0.149
	Average tradable share	0.400	0.400	0.400
	Std of tradable share	0.042	n.a.	0.042

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